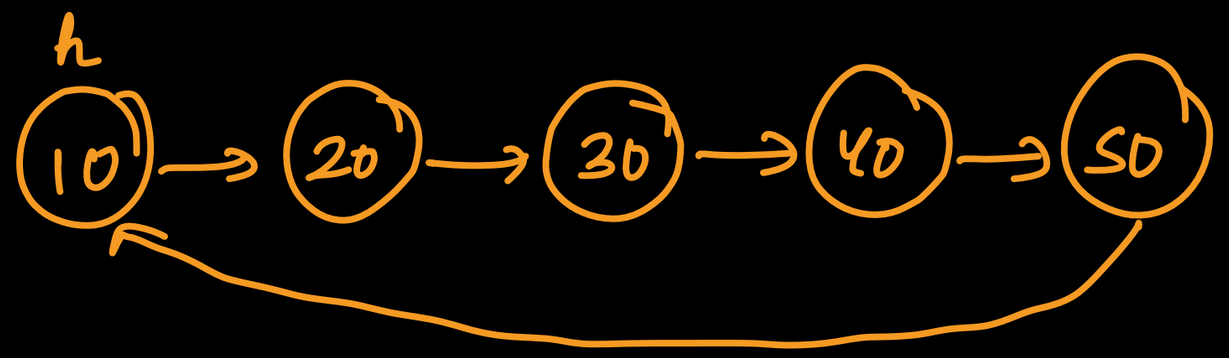
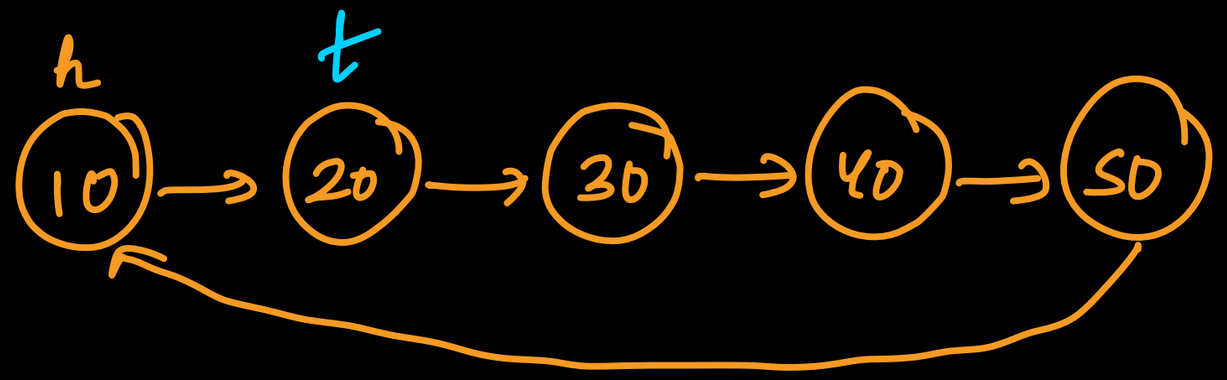
Circular Linked List



Ques: Circular Linked List Traversal



10
20
30
40
50

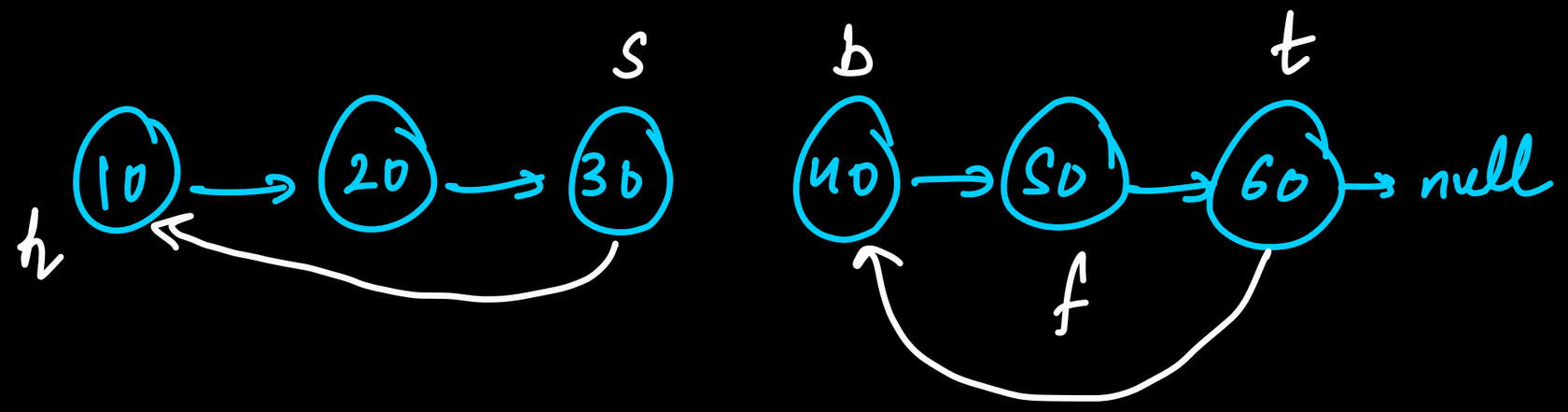
```
while (temp.next != head){
    sout (temp.val);
    temp = temp.next;
}
sout (temp.val);
```

GFG Practice

HW: Convert Singly LL to Circular LL

GFG Practice

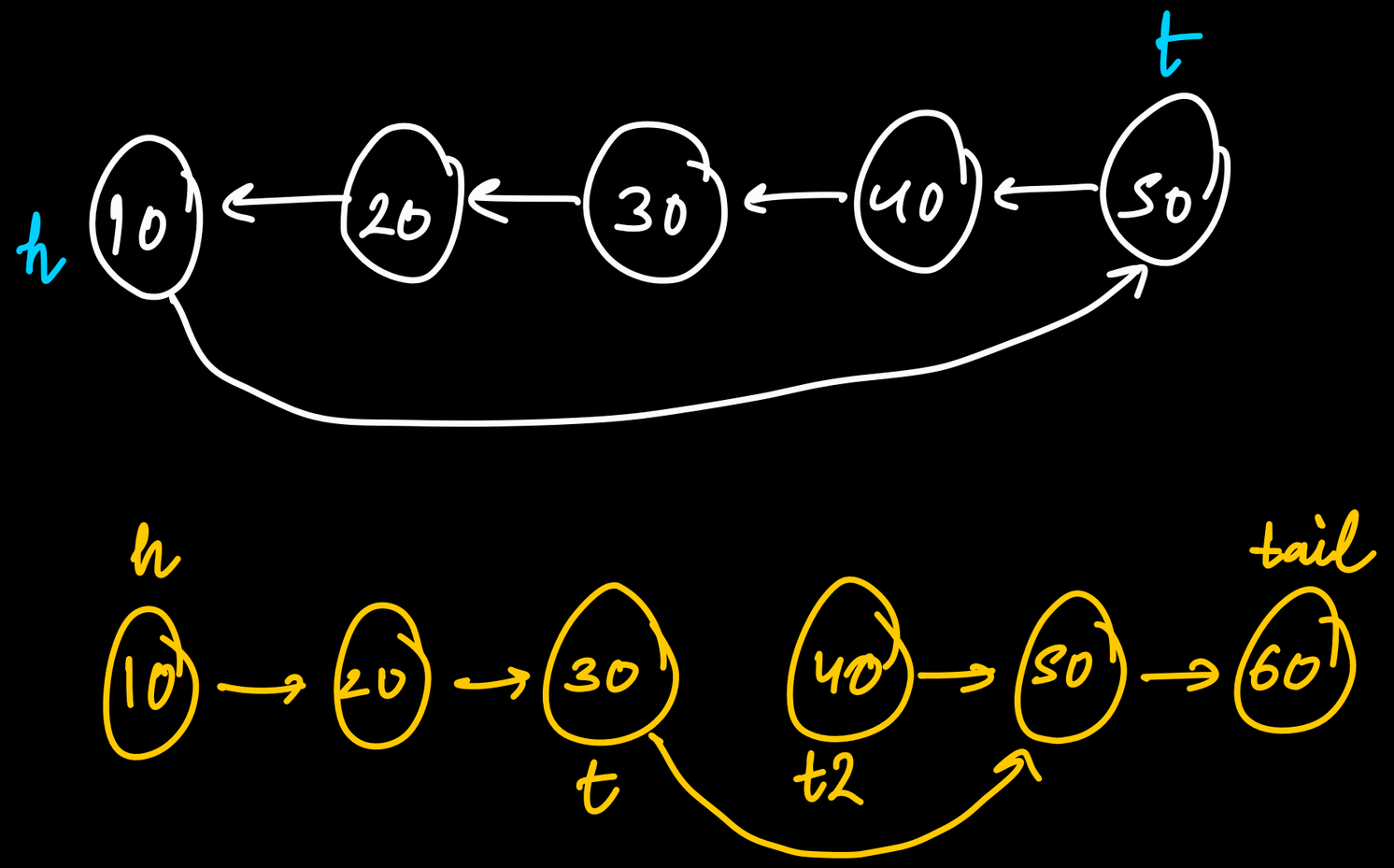
HW: Split a Linked List into 2 halves



Pair < Node, Node >

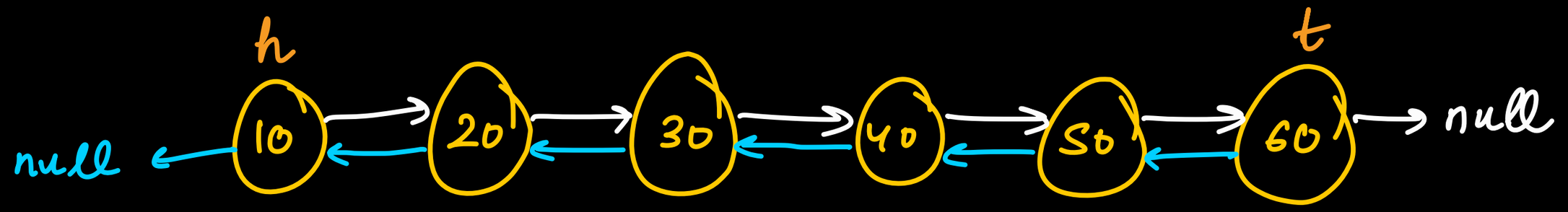
GFG Practice

Ques: Deletion and Reverse in Circular Linked List



GFG Practice

Doubly Linked List

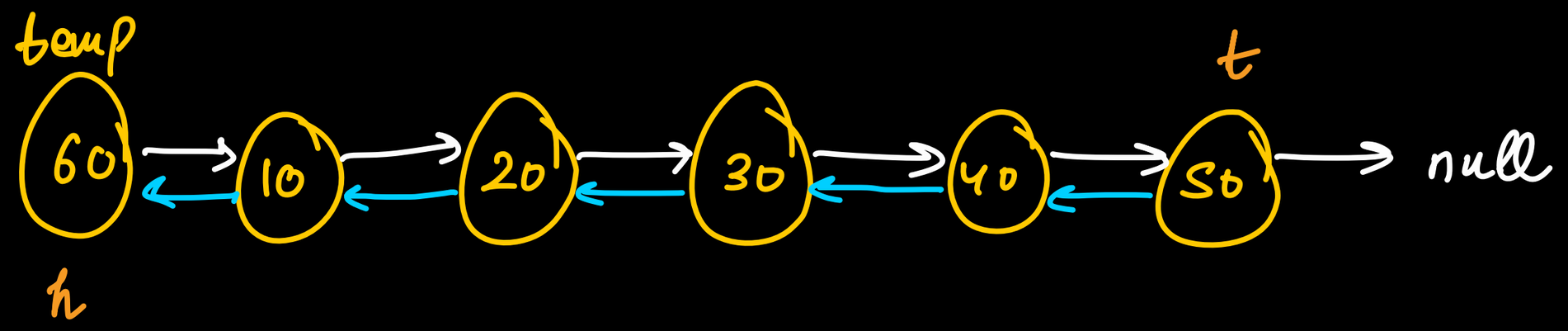



```
class Node {
    int val;
    Node next;
    Node prev;
}
```

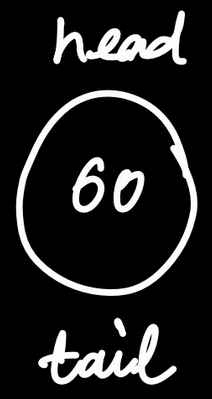
Doubly Linked List Class

```
class DLL {
    Node head;
    Node tail;
    int size;

    addAtHead
    addAtTail
    deleteAtHead
    deleteAtTail
    display
    displayReverse
    insert
    delete

}
```

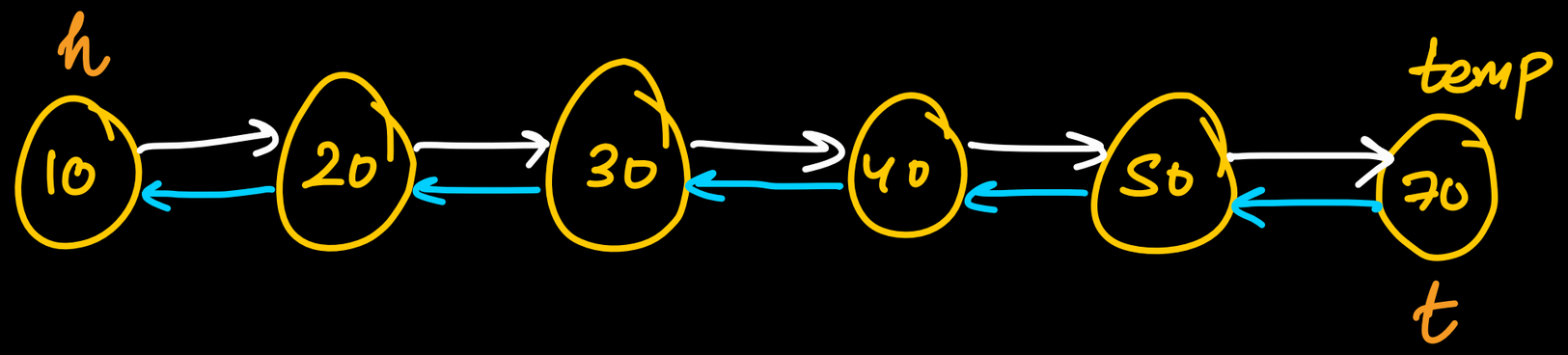
addAtHead(val)



Empty



addAtTail(val)

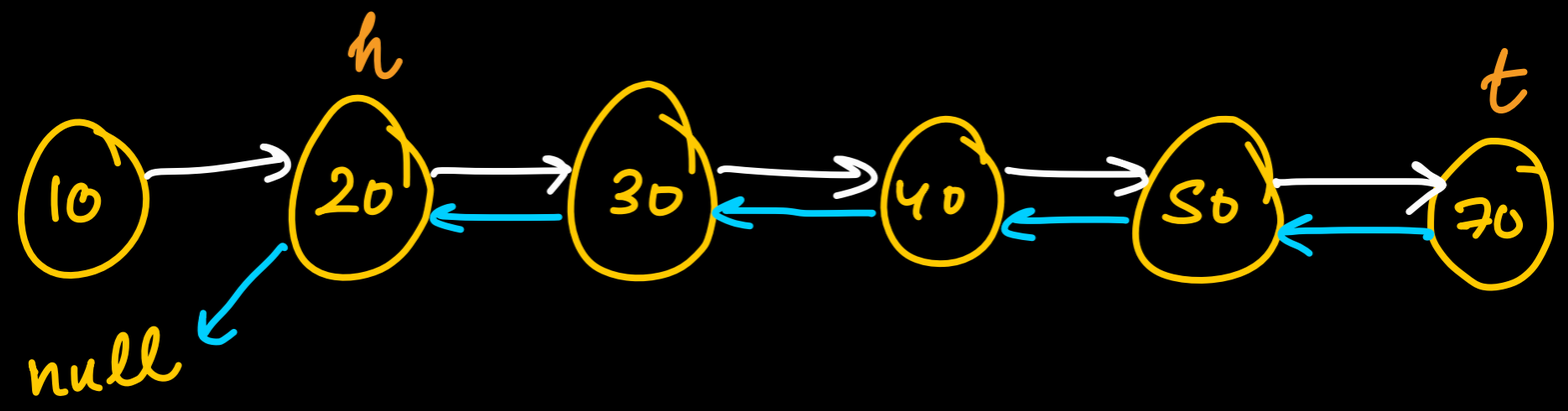


tail.next = temp

temp.prev = tail

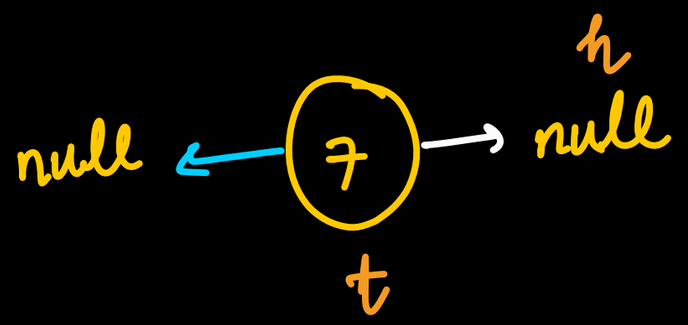
tail = temp

deleteAtHead()

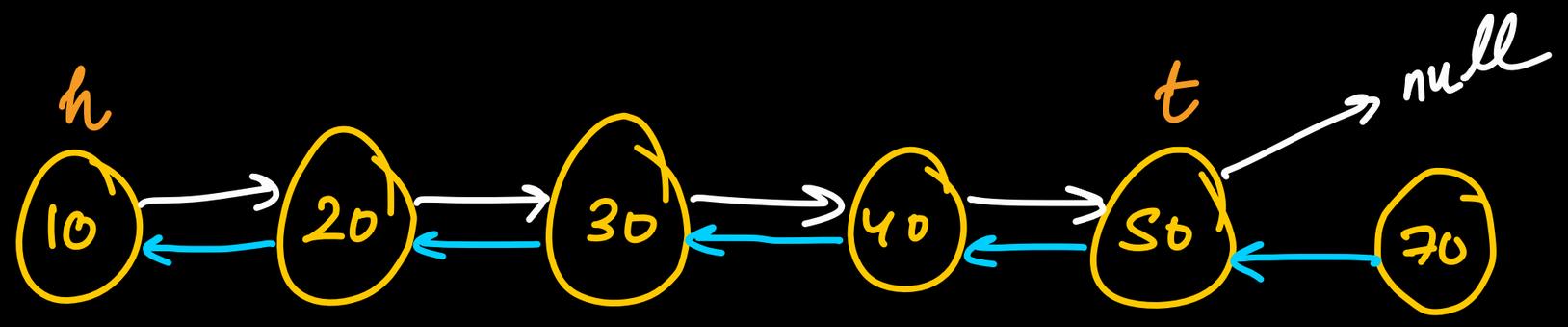


head = head.next

head.prev = null

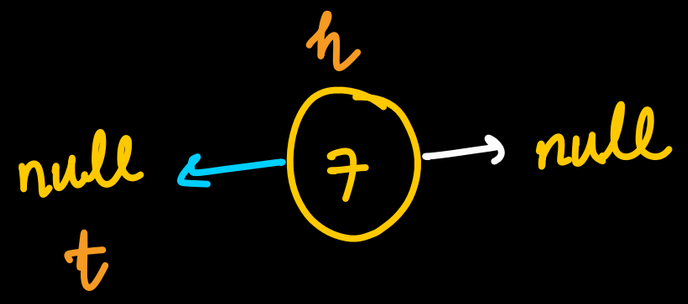


deleteAtTail()

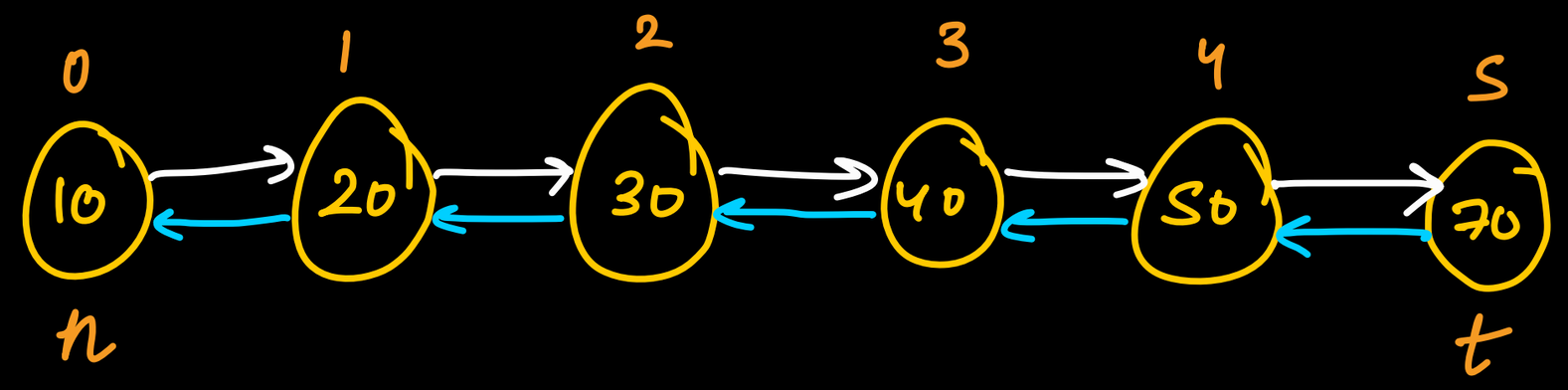


tail = tail.prev

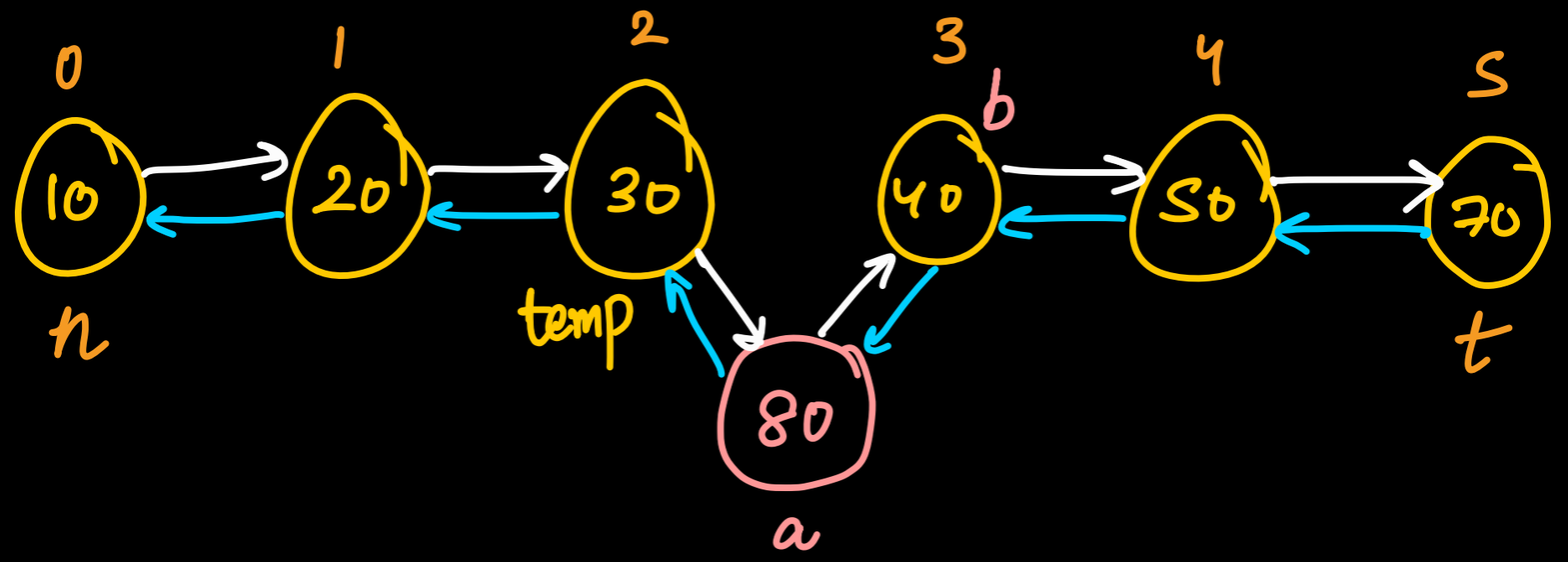
tail.next = null



insert (idx, val)

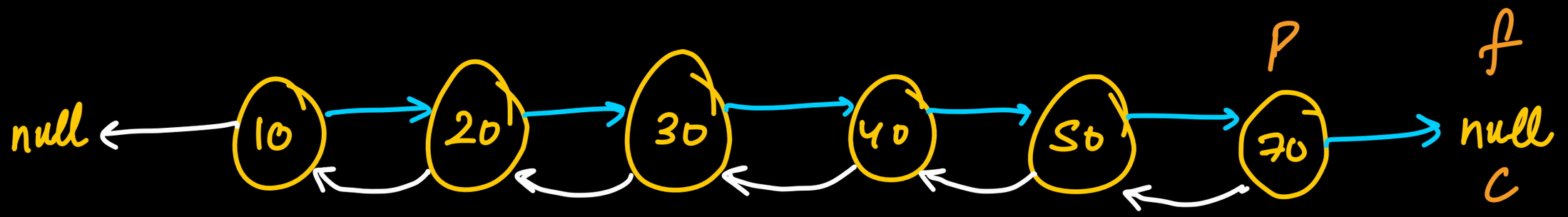


insert(3, 80);

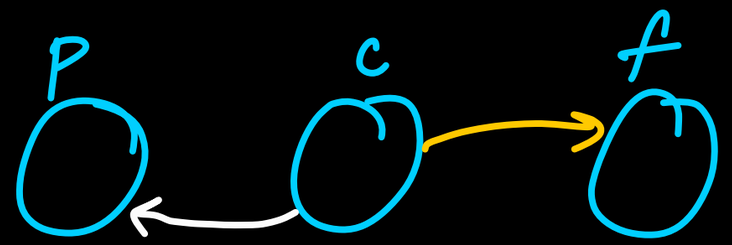


delete (idx)

Ques: Reverse a Doubly Linked List

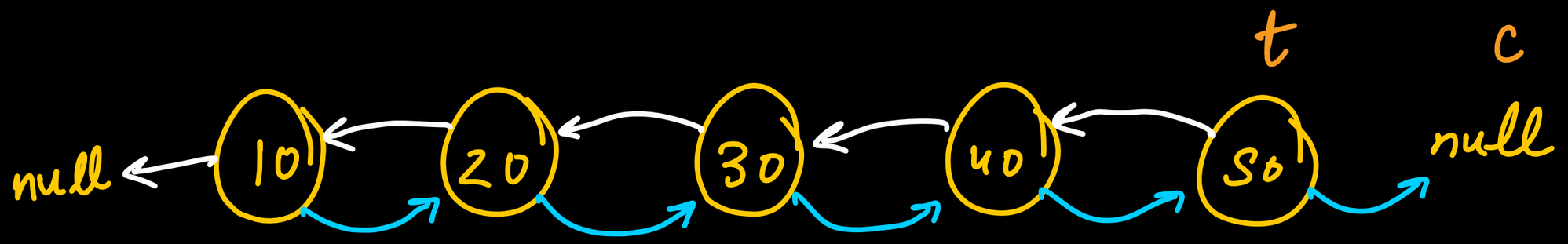



```
f = c.next
c.next = p
c.prev = f
p = c
c = f
```



GFG Practice

Ques: Reverse a Doubly Linked List



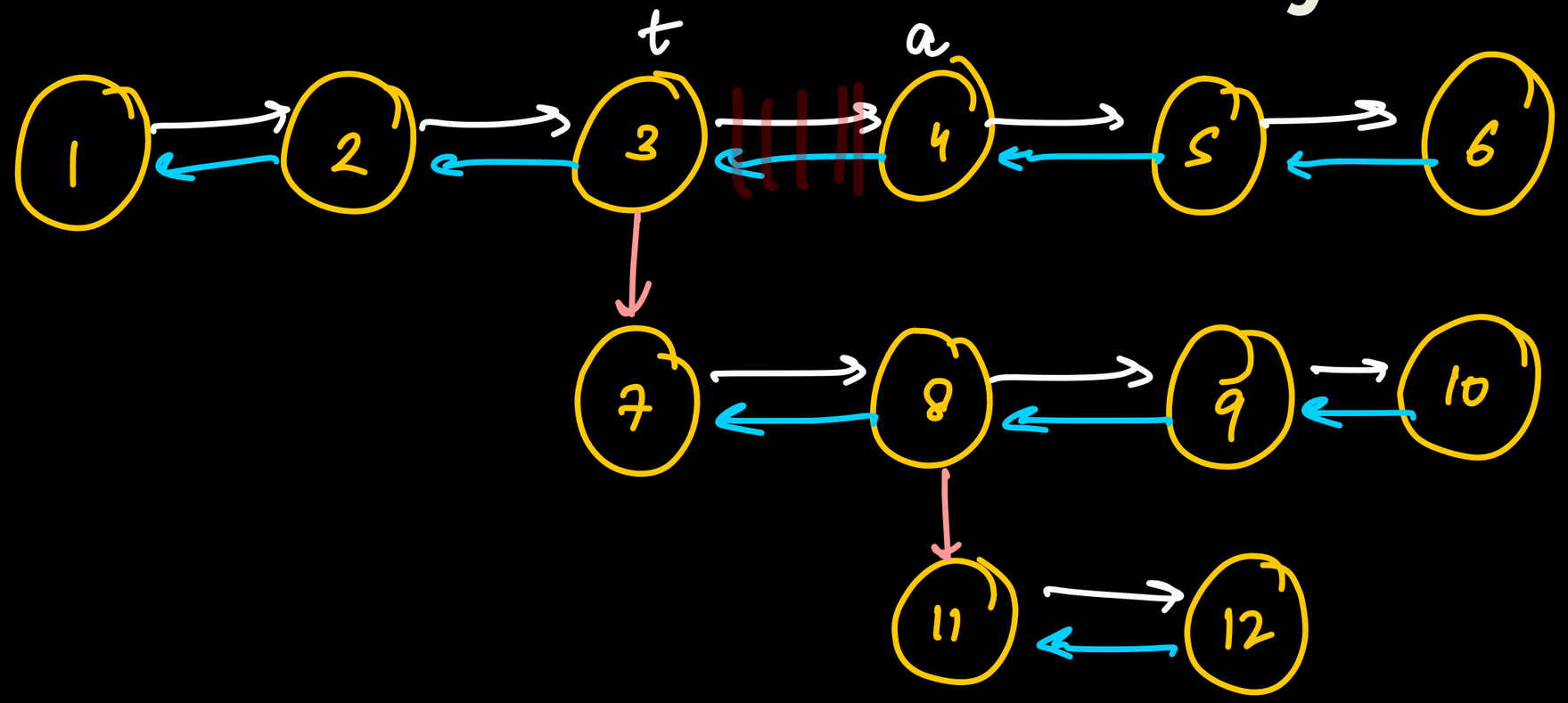

```
loop {
    t = c.prev
    c.prev = c.next
    c.next = t
    c = c.prev
}
t = t.prev
```

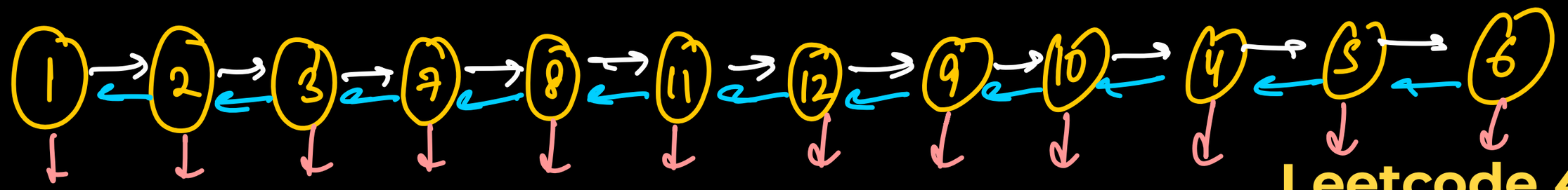
GFG Practice

HW: Remove Duplicates from Sorted Doubly Linked List

GFG Practice

Ques: Flatten a Multilevel Doubly Linked List





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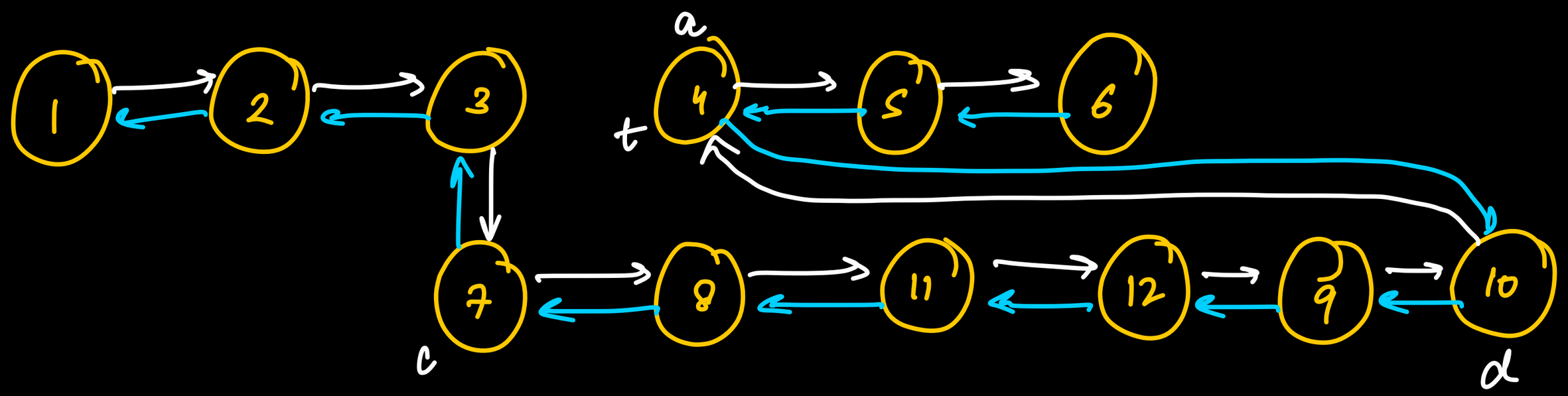
Ques: Flatten a Multilevel Doubly Linked List



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HW: Rotate Doubly Linked List

GFG Practice